

3

Level 3 Calculus

91578 Apply differentiation methods in solving problems

PRACTICE ASSESSMENT A (v10)

Achievement	Achievement with Merit	Achievement with Excellence
Apply differentiation methods in solving problems.	Apply differentiation methods, using relational thinking, in solving problems.	Apply differentiation methods, using extended abstract thinking, in solving problems.

You should attempt ALL the questions in this booklet.

Make sure that you have the Formulae and Tables Booklet L3–CALCF.

Show ALL working.

If you need more room for any answer, use the extra space provided at the back of this booklet.

Check that this booklet has pages 2–13 in the correct order and that none of these pages is blank.

QUESTION ONE

- (a) Differentiate $f(x) = \sqrt[3]{(3x - 2)^2}$.

You do not need to simplify your answer.

- (b) Find the gradient of the tangent to the curve $y = (2x^2 - 5x + 3) \cos(4x)$ at the point where $x = \frac{\pi}{4}$.

You must use calculus and show any derivatives that you need to find when solving this problem.

- (c) Consider the function $f(x) = \frac{x^2 - 2x - 3}{x - 1}$, where $x > 1$. A normal to the curve is parallel to the line $y = -\frac{1}{2}x + 8$.

Find the x -coordinate of the point where the curve has this normal.

You must use calculus and show any derivatives that you need to find when solving this problem.

(d) Consider the curve defined by the pair of parametric equations:

$x = t^2 \ln(2t)$ and $y = 2t^2 - t$, where $t > 0$.

Find the coordinates of the point where the tangent to the curve at its stationary point intersects the tangent at the point where the curve crosses the y -axis.

You must use calculus and show any derivatives that you need to find when solving this problem.

Question One continues
on the next page.

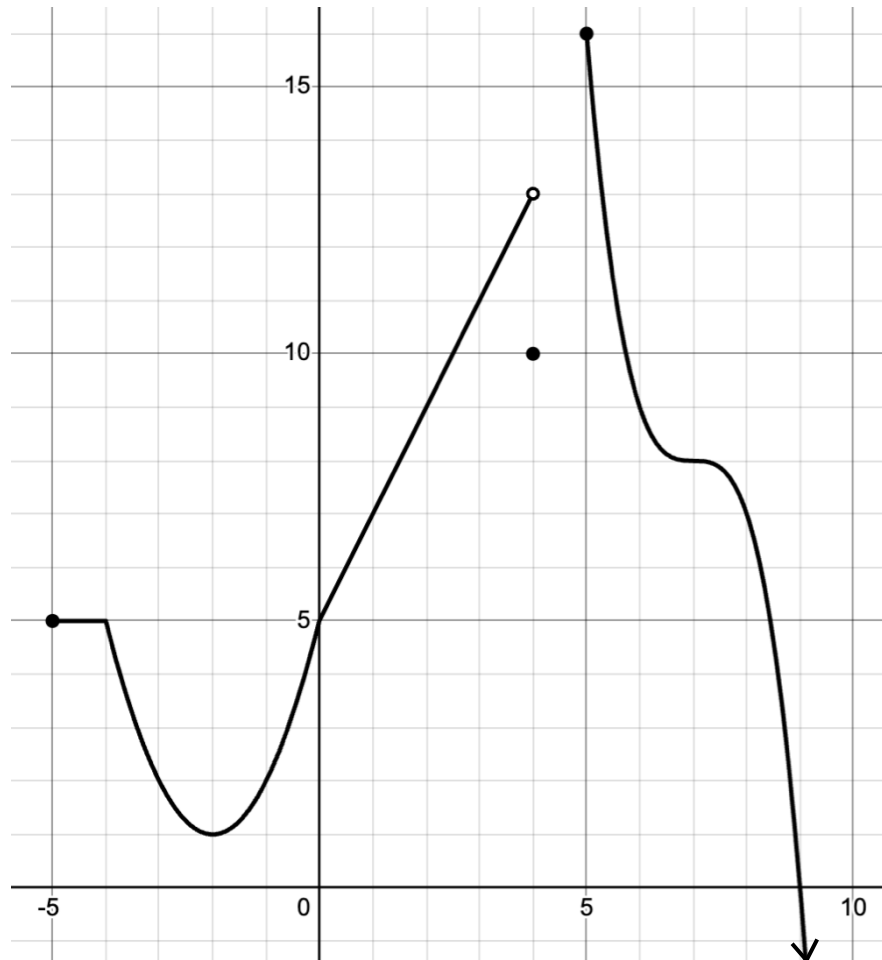
(e) Consider the function $f(x) = \tan(3x^2) - 3x^2$. A line L is normal to the graph of $y = f'(x)$.

Show that the line L is parallel to the y -axis at $x = \sqrt{\frac{\pi}{3}}$.

You must use calculus and show any derivatives that you need to find when solving this problem.

QUESTION TWO

(a) The graph below shows the function $y = f(x)$.



For the function above:

(i) Find the value(s) of x where $f'(x) = 0$.

(ii) Find the value(s) of x where $f(x)$ is concave upwards.

(iii) Find the value of $\lim_{x \rightarrow 4} f(x)$. State clearly if the value does not exist.

- (b) Show whether the function $f(x) = \frac{e^{2x}}{(x+2)^2}$ ($x \neq -2$) is increasing or decreasing at the point where $x = 0$.

You must use calculus and show any derivatives that you need to find when solving this problem.

- (c) A curve is defined by the pair of parametric equations: $x = t^3 + 3t$ and $y = t^6 - 3t^4 - 9t^2$.

Find the coordinate(s) of any point(s) of inflection on this curve.

You may assume that any point(s) of inflection found is/are actually a point of inflection.

You must use calculus and show any derivatives that you need to find when solving this problem.

(d) Given that $y = e^{3x} \sin x$ is a solution to the equation $\frac{d^2y}{dx^2} + ky = 6\frac{dy}{dx}$, find the value of the constant k .

Show ALL working clearly.

You must use calculus and show any derivatives that you need to find when solving this problem.

*Question Two continues
on the next page.*

(e) Consider the curve $y = \ln(e^x + e^{-x})$.

Explain whether the curve has any point(s) of inflection AND account for the concavity of the curve.

You must use calculus and show any derivatives that you need to find when solving this problem.

QUESTION THREE

(a) Differentiate $f(x) = 2x^2 \cot(x^3)$.

You do not need to simplify your answer.

(b) In an experiment, the temperature of a chemical is given by the formula $T = \frac{30t}{\sqrt{e^3}}$, where T is the temperature in $^{\circ}\text{C}$, t is the time after the experiment starts in minutes, and $0 \leq t \leq 40$.

Find the **exact time** t at which the temperature reaches a turning point.

You do not need to determine the nature of the turning point.

You must use calculus and show any derivatives that you need to find when solving this problem.

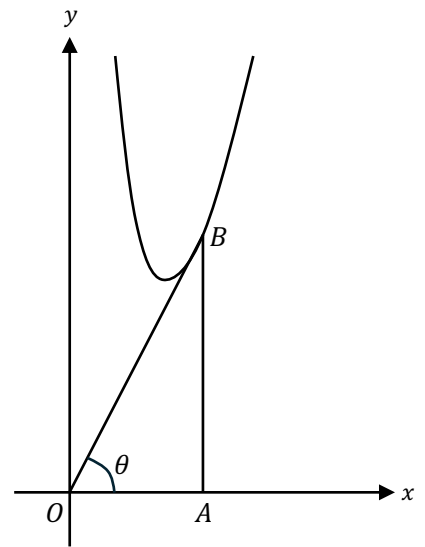
*Question Three continues
on the next page.*

- (c) A right-angled triangle OAB is formed in the first quadrant. Vertex O is fixed at the origin $(0, 0)$. Vertex A lies on the positive x -axis at $(x, 0)$. Vertex B lies vertically above A on the curve defined by the equation $y = x^2 + \frac{4}{x}$.

Let θ be the angle AOB inside the triangle at the origin. Find the x -coordinate of point A that minimises the angle θ AND the size of the minimum angle θ .

You may assume that the angle you have found is a minimum.

You must use calculus and show any derivatives that you need to find when solving this problem.



- (d) An expandable cylindrical container is being filled with water at a constant rate of $12\pi \text{ m}^3/\text{min}$.

The container is designed such that its height h is always exactly double its radius r .

Find the rate at which the surface area of the container is increasing at the instant when the radius is 4 metres.

You must use calculus and show any derivatives that you need to find when solving this problem.

*Question Three continues
on the next page.*

- After Sam takes off at point A , he climbs in a straight line directly to point B on PT . After reaching point B , the plane levels off and flies horizontally through an air corridor to point C on TQ . From point C , it descends in a straight line directly to point D .

Find the minimum flight time from point A to point D .

